For patient i, let G_i be the nonempty binary reference mask and $B_i = \left(x_i^-, y_i^-, x_i^+, y_i^+\right)$ its tight bounding box. Upper coordinates are one pixel beyond the largest foreground index, so $w_i = x_i^+ - x_i^-$ and $h_i = y_i^+ - y_i^-$. The initial prompt expands each side by 10 % and clips coordinates to the image (Eq. (1)):

$$\begin{aligned} B_{0,i} &= \mathrm{clip}_{\Omega_i}\left(B_i + e_i\right), \\ e_i &= \left(-0.10\, w_i, -0.10\, h_i, 0.10\, w_i, 0.10\, h_i\right). \end{aligned} \tag{1}$$

Here clip_{Ω_i} clips horizontal coordinates to $\left[0, W_i\right]$ and vertical coordinates to $\left[0, H_i\right]$, where W_i, H_i are the image dimensions. Writing the clipped box width and height as $w_{0,i}, h_{0,i}$, its four perturbations are

$$\begin{aligned} B_{j,i} &= \mathrm{clip}_{\Omega_i}\left\{B_{0,i} + \left(u_{j,i}, v_{j,i}, u_{j,i}, v_{j,i}\right)\right\}, \\ \left(u_{j,i}, v_{j,i}\right) &\in \left\{\left(-\delta_{x,i}, 0\right), \left(\delta_{x,i}, 0\right), \left(0, -\delta_{y,i}\right), \left(0, \delta_{y,i}\right)\right\}, \\ \delta_{x,i} &= 0.05\, w_{0,i}, \delta_{y,i} = 0.05\, h_{0,i}. \end{aligned} \tag{2}$$

The four entries in Eq. (2) correspond to left, right, up and down, respectively. This reference-informed, known-location (oracle-localization) setting does not test autonomous lesion localization. We encoded the image once and reused its embedding for the initial and four shifted-prompt predictions (Figure 2). The same 10 % context margin and 5 % shifts were used across datasets.

Let $P_{0,i}$ denote the initial binary prediction and $P_{j,i}$ its prediction under shift j. For two binary masks U, V, with $|\cdot|$ denoting foreground-pixel count, Dice agreement is

$$D(U,V)=\begin{cases}\dfrac{2|U\cap V|}{|U|+|V|}, & |U|+|V|>0,\\ 1, & |U|+|V|=0.\end{cases} \tag{3}$$

Method A uses the raw predicted intersection over union (IoU), $\hat{q}_i$, returned by the SAM quality head for the initial mask:

$$r_{A,i}=1-\hat{q}_i. \tag{4}$$

Method B uses disagreement between the initial prediction and the four shifted-prompt predictions:

$$r_{B,i}=1-\frac{1}{4}\sum_{j=1}^{4} D\left(P_{0,i},P_{j,i}\right). \tag{5}$$

Thus $\hat{q}_i$ in Eq. (4) is a model output, whereas Eq. (5) measures prediction-to-prediction consistency. All four directions were retained when clipping produced coincident boxes. Method B adds four mask-decoder passes while reusing the image embedding. Recorded inference times cover the combined pipeline; A-only and incremental B times were not separately benchmarked.

The quality reference for both methods was $D\left(P_{0,i},G_i\right)$, separate from the scores in Eqs. (4) and (5). We defined the low-quality indicator as

$$Y_i=1\{D\left(P_{0,i},G_i\right)<\tau\},\tau=0.70. \tag{6}$$

Here $1\{\cdot\}$ is one when its condition holds and zero otherwise. This is an operational quality cutoff. An empty initial prediction against the nonempty reference has Dice zero under Eq. (3).

OpenAI Codex assisted with study-design discussions and implementation of research, statistical and plotting code. The pipeline processed public images with frozen SAM weights, and figures were rendered conventionally from recorded data and captured predictions. The human authors retain responsibility for the implementation and calculated outputs. A precise underlying Codex model identifier was not available in the archived study records.

Development threshold selection

For method $m \in \{A, B\}$, candidate cutoff t flags patient i when the risk meets or exceeds the cutoff. Its development review fraction is given by Eq. (7):

$$\begin{aligned} a_{m,i}(t) &= 1\{r_{m,i} \geq t\}, \\ v_{m,\mathrm{dev}}(t) &= \frac{1}{N_{\mathrm{dev}}} \sum_{i \in \mathrm{dev}} a_{m,i}(t). \end{aligned} \tag{7}$$

The candidate set T_m contains every distinct development risk and a cutoff above the maximum risk, representing zero reviews. For target fraction b, calibration selects

$$t_m^* = \operatorname*{argmin}_{t \in T_m}{}_{\mathrm{lex}} \left(|v_{m,\mathrm{dev}}(t) - b|, v_{m,\mathrm{dev}}(t) \right). \tag{8}$$

The ordered pair in Eq. (8) is minimized lexicographically: first the absolute target discrepancy, then the review fraction. Thus cutoff ties are kept together and equally close choices select fewer cases. The primary target was $b = 0.20$, a prespecified one-in-five scenario rather than measured clinical capacity. The exact thresholds were locked before test scoring and transferred unchanged to both test cohorts.

Outcomes and statistical analysis

The analysis included cases with valid reference Dice and both finite risk scores. One retry with the same configuration was allowed after a technical failure; persistent failures were recorded separately. In a cohort of N scored patients, let $F = \sum_i Y_i$ be the number of low-quality masks. Writing $a_{m,i}$ for the flag under the evaluated policy, the counts are $\mathrm{TP}_m = \sum_i Y_i a_{m,i}$, $\mathrm{FP}_m = \sum_i (1 - Y_i) a_{m,i}$, $\mathrm{FN}_m = F - \mathrm{TP}_m$ and $\mathrm{TN}_m = N - F - \mathrm{FP}_m$. These count flagged low-quality, flagged acceptable, unflagged low-quality and unflagged acceptable masks, respectively. The reported outcomes are

$$\begin{aligned} d_m &= \frac{\mathrm{TP}_m}{F}, v_m = \frac{\mathrm{TP}_m + \mathrm{FP}_m}{N}, \\ M_m &= 100 \frac{\mathrm{FN}_m}{N}, F = \mathrm{TP}_m + \mathrm{FN}_m. \end{aligned} \tag{9}$$

Here d_m is low-quality detection, v_m is simulated review fraction and M_m is missed low-quality masks per 100 patients. Rates with a zero denominator were left undefined.

Let E and I denote BrEaST external testing and BUS-BRA internal testing. The dated plan specified the following differences in percentage points (Eq. (10)):

$$\begin{aligned}\Delta_{\mathrm{H1}} &= 100\left(d_{A,E} - d_{A,I}\right),\\ \Delta_{\mathrm{H2}} &= 100\left(v_{A,E} - v_{A,I}\right),\\ \Delta_{\mathrm{H3}} &= 100\left(d_{B,E} - d_{A,E}\right).\end{aligned} \tag{10}$$

H1 was primary; H2 assessed the corresponding workload difference. H3 compared the scores on the same external low-quality masks at separately locked cutoffs that gave equal development review counts. We calculated two-sided 95 % confidence intervals (CIs) using Newcombe score method 10 for the independent proportions in H1 and H2 [11] and its paired counterpart in H3 [12]. H2, H3 and supplementary intervals are nominal, with no multiplicity adjustment. The exact interval equations are given in the Supplementary mathematical methods.

Sensitivity analyses replaced τ in Eq. (6) by 0.50 or 0.80 while retaining the locked risk thresholds. A separate equal-review policy selects the same nearest-integer count for each method:

$$\begin{aligned}K(b,N) &= \min \operatorname*{argmin}_{k \in \{0,\ldots,N\}} |k - bN|,\\ a_{m,i}^{(K)} &= 1\{\mathrm{rank}_m(i) \le K(b,N)\}.\end{aligned} \tag{11}$$

The minimum in Eq. (11) resolves an exact half-integer tie in favor of fewer reviews. Rank is determined by decreasing risk, then the unmodified archived patient-identifier string in lexicographic order, for example “1”, “10”, “100”. With $b = 0.20$ and $N = 252$, each method selects the top 50 external cases. This paired interval conditions on the observed rankings.

Post hoc descriptive quality-control measures used low-quality segmentation as the positive condition and a review flag as the positive test. Sensitivity equals d_m in Eq. (9); the remaining measures are given by Eq. (12):

$$\begin{aligned} \text{Specificity}_m &= \frac{\text{TN}_m}{\text{TN}_m + \text{FP}_m}, \\ \text{PPV}_m &= \frac{\text{TP}_m}{\text{TP}_m + \text{FP}_m}, \\ \text{NPV}_m &= \frac{\text{TN}_m}{\text{TN}_m + \text{FN}_m}. \end{aligned} \tag{12}$$

PPV and NPV denote positive and negative predictive value. We reported the TP, FP, TN and FN counts and Wilson 95 % intervals for these proportions. These measures assess mask-quality screening, not diagnosis of breast cancer; the Wilson equation is provided in the Supplementary mathematical methods.

We additionally performed a post hoc exploratory stratification by dataset-reported benign/malignant classification, retaining the locked risk thresholds and Dice < 0.70 definition. BUS-BRA labels came from the official Pathology field, linked by image and patient identifiers. BrEaST labels came from Classification, linked by CaseID and verified against the selected filenames; 197 diagnoses were biopsy-confirmed and 55 were confirmed by follow-up. We report stratum counts, review fractions and detection with Wilson 95 % intervals. No stratum-specific recalibration, hypothesis test or interaction test was performed. We also audited all external low-quality cases missed by A, partitioned by the B review flag.

Post hoc policy sensitivity and diagnostic accounting

The following exploratory analyses were specified for this revision after the original study and retain its scores, low-quality definition and H1–H3. We evaluated $b \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50\}$. For each method and target, Eq. (8) selects a threshold using development risks and counts without quality labels; the resulting numerical cutoff is transferred unchanged to both test cohorts. Separately, Eq. (11) selects a cohort-specific equal-review count and ranking.

We used 10,000 patient-bootstrap replicates per route (seed 20261106) to separate test-cohort sampling at the original cutoffs, development resampling and recalibration with fixed test cohorts, and both together. Cohorts were sampled independently, keeping A/B scores and quality labels paired within each record. Repeated draws were retained. Equal review selects

exactly 50 external draw records, ordered by decreasing risk, original identifier string and draw-position index. The strict original direction pattern requires external B-minus-A detection above zero at transferred cutoffs and below zero at equal review; ties are separate. Bootstrap percentile 95 % intervals and direction-frequency equations are given in the Supplementary mathematical methods. These are descriptive, nominal sensitivity analyses and do not replace the prespecified Newcombe intervals.

We also used symmetric Kitagawa accounting and direct standardization across dataset-reported benign/malignant strata [13]. Let s denote a diagnostic stratum, $w_{c,s}$ its weight and $p_{c,s}$ its outcome rate in cohort $c \in \{E, I\}$. When all stratum rates are defined, $P_c = \sum_s w_{c,s} p_{c,s}$ and the external-minus-internal difference is given by Eq. (13):

$$\begin{aligned} \Delta_P = P_E - P_I &= C + W, \\ C &= \sum_s \left(w_{E,s} - w_{I,s}\right) \frac{p_{E,s} + p_{I,s}}{2}, \\ W &= \sum_s \left(p_{E,s} - p_{I,s}\right) \frac{w_{E,s} + w_{I,s}}{2}. \end{aligned} \tag{13}$$

Here C is the composition-associated term and W the within-stratum-rate term. Common weights give directly standardized rates (Eq. (14)):

$$\begin{aligned} w_s^* &= \frac{w_{E,s} + w_{I,s}}{2}, P_c^{\text{std}} = \sum_s w_s^* p_{c,s}, \\ P_E^{\text{std}} - P_I^{\text{std}} &= W. \end{aligned} \tag{14}$$

For detection, $w_{c,s} = F_{c,s} / F_c$ and $p_{c,s} = \text{TP}_{c,s} / F_{c,s}$; for review, $w_{c,s} = N_{c,s} / N_c$ and $p_{c,s} = R_{c,s} / N_{c,s}$, where $R_{c,s}$ is the flagged count. We obtained nominal percentile intervals from 5,000 independent cohort-level patient-bootstrap replicates (seed 20261107), recalculating weights and rates. Undefined stratum rates remained missing and propagated to dependent components, including when the associated weight was zero. Overall rates were computed directly from aggregate numerators and denominators. This accounting describes observed mixtures and does not estimate causal effects.

Computational reproducibility

The original study assessed computational reproducibility by recomputing statistics from archived case scores and by rerunning all 1,316 images in the original computing environment. Its recorded comparisons covered eight numerical fields, quality labels, review

Supplementary mathematical methods

These equations specify the interval calculations and post hoc analyses used in this study. Definitions of the masks, risk scores, low-quality condition and review policies are given in the main Methods. Rates below are fractions unless explicitly multiplied by 100; interval limits for differences are reported in percentage points. Citations [11–13] refer to the main reference list.

Proportion and difference intervals

For x successes among n observations, let $\hat{p} = x / n$ and $z = \Phi^{-1}(0.975)$. The two-sided 95 % Wilson score limits without continuity correction are given by Eq. (S1):

$$(L, U) = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \tag{S1}$$

The minus sign gives L and the plus sign U. The implementation clips these proportion limits to [0,1].

Here x and n are the numerator and denominator of the reported proportion. Detection uses low-quality patients as its denominator; review uses all scored patients. A proportion with a zero denominator is undefined and is not assigned an interval.

For the first-minus-second contrast $\hat{\Delta} = \hat{p}_1 - \hat{p}_2$, let (L_j, U_j) be the Wilson limits of $\hat{p}_j$. The independent or paired Newcombe method 10 interval [11,12] is given by Eq. (S2):

$$\begin{aligned} R(x, y; \rho) &= \sqrt{\max(x^2 + y^2 - 2\rho x y, 0)}, \\ L_\Delta &= \hat{\Delta} - R(\hat{p}_1 - L_1, U_2 - \hat{p}_2; \rho), \\ U_\Delta &= \hat{\Delta} + R(U_1 - \hat{p}_1, \hat{p}_2 - L_2; \rho). \end{aligned} \tag{S2}$$

Set $\rho = 0$ for the independent external-minus-internal H1 and H2 contrasts. For paired comparisons, define a, b, c and d as the numbers flagged by both methods, by the first method only, by the second method only, and by neither method, within the population used for the contrast. That population comprises low-quality patients for detection and all scored patients for review. For a B-minus-A contrast, first therefore denotes B. The paired method uses Eqs. (S3) and (S4):

$$N = a + b + c + d,\ T = a\,d - b\,c,\ M = (a+b)(c+d)(a+c)(b+d). \tag{S3}$$

$$T^{\mathrm{corr}} = T - 1\{T > 0\}\min(T, N/2),\ \rho = \frac{T^{\mathrm{corr}}}{\sqrt{M}}\,(M > 0). \tag{S4}$$

If $M = 0$, set $\rho = 0$. This correction acts only on positive association; a negative T and its negative correlation are retained. The maximum with zero in R handles numerical round-off. H1 uses the original independent interval; H2, H3 and supplementary intervals are nominal without multiplicity adjustment. Equal-review paired intervals condition on the observed selections and do not include calibration or re-ranking uncertainty.

Patient-bootstrap summaries

Let $\theta^{*(b)}$ denote the statistic in replicate b, and V the set of replicates for which that statistic is finite. The descriptive percentile interval is given by Eq. (S5):

$$\mathrm{CI}^{\mathrm{boot}}_{95\,\%} = \left[Q_{0.025}\{\theta^{*(b)} : b \in V\}, Q_{0.975}\{\theta^{*(b)} : b \in V\}\right]. \tag{S5}$$

Quantiles use linear interpolation of the sorted finite replicate values. Undefined detection denominators remain missing; no values are imputed. The three calibration/sampling routes each use 10,000 patient-bootstrap

replicates. Within each sampled patient record, A/B scores and the quality label remain paired; repeated patient draws are retained. Cohorts are resampled independently.

For joint direction summaries, let V_J contain replicates with both external B-minus-A detection contrasts defined: Δ^*_{fixed} at transferred thresholds and Δ^*_{equal} at equal top-50 review. The four mutually exclusive empirical frequencies are given by Eqs. (S6) and (S7):

$$\begin{aligned}\hat{f}_{+,-} &= \frac{1}{|V_J|}\sum_{b\in V_J} 1\{\Delta^{*(b)}_{\text{fixed}}>0, \Delta^{*(b)}_{\text{equal}}<0\},\\ \hat{f}_{-,+} &= \frac{1}{|V_J|}\sum_{b\in V_J} 1\{\Delta^{*(b)}_{\text{fixed}}<0, \Delta^{*(b)}_{\text{equal}}>0\}.\end{aligned} \tag{S6}$$

$$\begin{aligned}\hat{f}_{0} &= \frac{1}{|V_J|}\sum_{b\in V_J} 1\{\Delta^{*(b)}_{\text{fixed}}=0 \vee \Delta^{*(b)}_{\text{equal}}=0\},\\ \hat{f}_{\text{same}} &= \frac{1}{|V_J|}\sum_{b\in V_J} 1\{\Delta^{*(b)}_{\text{fixed}}\Delta^{*(b)}_{\text{equal}}>0\}.\end{aligned} \tag{S7}$$

The original strict pattern is $(+,-)$, the opposite strict pattern is $(-,+)$, and the frequency of any strict reversal is $\hat{f}_{+,-}+\hat{f}_{-,+}$. A zero contrast is not a strict reversal. These frequencies describe resampling of the observed patients and are not p-values, posterior probabilities or population probabilities.

At the locked thresholds, the 16 external low-quality patients comprise 10 detected by both methods, none by A only, three by B only and three by neither. R1 cannot create an A-only detection from this empirical support, so its fixed-threshold B-minus-A distribution is structurally nonnegative. Its percentile interval does not exclude a population advantage for A. Bootstrap intervals supplement, and do not replace, the prespecified Newcombe intervals.

Diagnostic-stratum accounting and standardization

For cohort $c \in \{I, E\}$ and dataset-reported diagnostic stratum s (benign or malignant), write $d_{c,s}$ for the denominator and $u_{c,s}$ for its numerator. In the detection analysis, $d_{c,s} = F_{c,s}$ (low-quality patients) and $u_{c,s} = \mathrm{TP}_{c,s}$ (flagged low-quality patients). In the review analysis, $d_{c,s} = N_{c,s}$ (all scored patients) and $u_{c,s} = R_{c,s}$ (flagged patients). The rates and weights are defined in Eq. (S8):

$$p_{c,s} = \frac{u_{c,s}}{d_{c,s}}, w_{c,s} = \frac{d_{c,s}}{\sum_t d_{c,t}}, P_c = \frac{\sum_s u_{c,s}}{\sum_s d_{c,s}}. \tag{S8}$$

When the stratum rates are defined, $P_c = \sum_s w_{c,s} p_{c,s}$. The symmetric Kitagawa identity [13] partitions the external-minus-internal difference into a composition-associated term C and a within-stratum-rate term W (Eqs. (S9) and (S10)):

$$\Delta_{EI} = P_E - P_I = C + W. \tag{S9}$$

$$\begin{aligned} C &= \sum_s \left(w_{E,s} - w_{I,s}\right) \frac{p_{E,s} + p_{I,s}}{2}, \\ W &= \sum_s \left(p_{E,s} - p_{I,s}\right) \frac{w_{E,s} + w_{I,s}}{2}. \end{aligned} \tag{S10}$$

Using the arithmetic-mean common weights gives direct-standardized cohort rates and a standardized difference equal to W (Eq. (S11)):

$$\begin{aligned} w_s^* &= \frac{w_{E,s} + w_{I,s}}{2}, P_c^{\mathrm{std}} = \sum_s w_s^* p_{c,s}, \\ P_E^{\mathrm{std}} - P_I^{\mathrm{std}} &= W. \end{aligned} \tag{S11}$$

All weights and stratum rates are recalculated in each of 5,000 within-cohort patient-bootstrap replicates. A stratum rate with a zero denominator remains undefined, even if its weight is zero; the missing value propagates to any component that uses that rate. Overall rates are calculated from aggregate numerators and denominators and remain defined when the total denominator is positive. This leaves 4,999 valid detection-component replicates because one draw has no external benign low-quality patient; detection totals and all review terms have 5,000 valid replicates. Each term uses its own finite-value percentile interval. Point components add to the total before rounding; interval endpoints need not add. This post hoc accounting is descriptive, not causal attribution or a percent-explained analysis, and two diagnostic strata do not remove other confounding.

Numerical replay differences

For the six selected patients, let v_{ij}^{CPU} and $v_{ij}^{\mathrm{archive}}$ be the CPU-replayed and archived values of numerical field j. The eight fields are reference Dice, initial predicted IoU, risk A, risk B and the four initial-to-shifted Dice values. We calculated signed differences and maximum absolute differences using Eq. (S12):

$$\begin{aligned} \delta_{ij} &= v_{ij}^{\mathrm{CPU}} - v_{ij}^{\mathrm{archive}}, \epsilon_j = \max_{i \in I_6} |\delta_{ij}|, \\ \epsilon_{\mathrm{all}} &= \max_{i \in I_6, j \in \{1,\ldots,8\}} |\delta_{ij}|. \end{aligned} \tag{S12}$$

Here I_6 is the same six-patient subset, so ϵ_{all} summarizes 48 numerical comparisons. Exact equality and agreement of the fixed review decisions were checked separately. Numerical agreement of selected scores or decisions does not establish pixel identity with the unarchived original CUDA masks, create independent patients, or demonstrate improved screening performance.